

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## UNIT 6 · FUNCTIONS · GUIDED NOTES

# Tables & Systems of Equations

*Fill in the blanks. Practice problems at the end.*

## 1 Writing Linear Equations from a Table

Every linear equation can be written as  $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$ , where  $m$  is the \_\_\_\_\_ (rate of change) and  $b$  is the \_\_\_\_\_ (the  $y$ -value when  $x = 0$ ).

$$m = (y_2 - y_1) \div (x_2 - x_1) = \text{change in } y \div \text{change in } x$$

**Example 1 · Table includes  $x = 0$** 

| x | y  |
|---|----|
| 0 | 3  |
| 1 | 7  |
| 2 | 11 |
| 3 | 15 |

**Step 1.** From (0, 3) to (1, 7):change in  $y = \underline{\hspace{1cm}}$ , change in  $x = \underline{\hspace{1cm}}$ slope  $m = \underline{\hspace{1cm}}$ **Step 2.** When  $x = 0$ ,  $y = \underline{\hspace{1cm}}$ . So  $b = \underline{\hspace{1cm}}$ .**Step 3.** Equation:  $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$ **Example 2 · No  $x = 0$  row — work backwards for  $b$** 

| x | y  |
|---|----|
| 2 | 5  |
| 4 | 11 |
| 6 | 17 |
| 8 | 23 |

**Step 1.** From (2, 5) to (4, 11):change in  $y = \underline{\hspace{1cm}}$ , change in  $x = \underline{\hspace{1cm}}$ .  $m = \underline{\hspace{1cm}}$ .**Step 2.** Use  $b = y - m \cdot x$ . Pick (2, 5): $b = \underline{\hspace{1cm}} - (\underline{\hspace{1cm}})(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$ **Step 3.** Equation:  $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$ **Shortcut for finding  $b$** Once you know  $m$ , pick any row ( $x, y$ ). Then:  $b = y - m \cdot x$

## 2 Solving Systems of Equations by Graphing

A system of equations is \_\_\_ or more equations that use the same \_\_\_\_\_. In 8th grade, we work with systems of two linear equations in  $x$  and  $y$ .

### Graphing Method

Graph each line on the same coordinate plane. The solution is the \_\_\_\_\_ point.

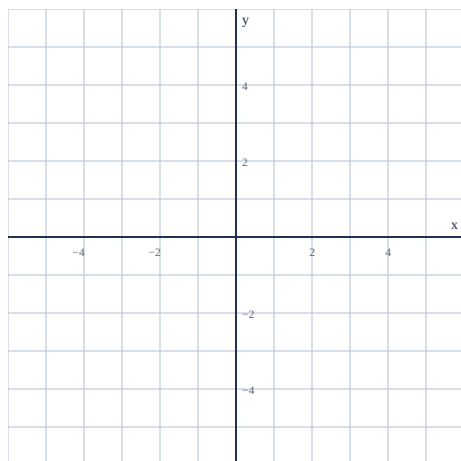
#### Steps

1. Graph the first line (using slope and  $y$ -intercept).
2. Graph the second line on the same plane.
3. Find where they \_\_\_\_\_. That ordered pair is the solution.
4. Check by plugging the point into both equations.

### Example · Graph and solve

EQ 1  $y = x + 2$

EQ 2  $y = -x + 4$



EQ 1:  $y = x + 2$

slope = \_\_\_,  $y$ -int = \_\_\_

EQ 2:  $y = -x + 4$

slope = \_\_\_,  $y$ -int = \_\_\_

**Solution (intersection):**

( \_\_\_, \_\_\_ )

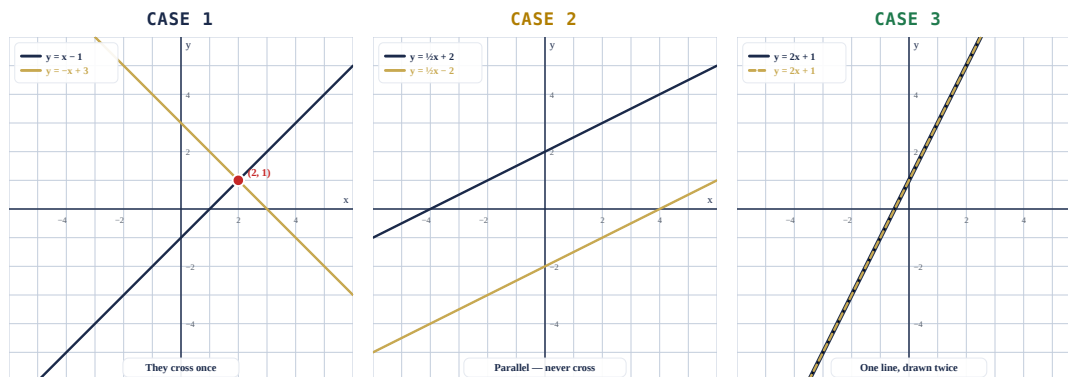
**Check:**

EQ 1: \_\_\_ = \_\_\_ + 2

EQ 2: \_\_\_ = -\_\_\_ + 4

### Three Possible Outcomes

| Case | What You See                       | How Many Solutions? |
|------|------------------------------------|---------------------|
| 1    | Lines cross at one point           | _____ solution      |
| 2    | Lines are _____ (never cross)      | _____ solutions     |
| 3    | Both equations graph the same line | _____ solutions     |



Count the intersection points: one point  $\rightarrow$  one solution, zero points  $\rightarrow$  no solution, every point  $\rightarrow$  infinitely many.

### 3 What Does “Solution” Actually Mean?

A solution to a system of equations is an ordered pair  $(x, y)$  that makes \_\_\_\_\_ equations \_\_\_\_\_ at the same time.

#### Two ways to say it

- **Graphically:** the solution is the \_\_\_\_\_ where the lines cross.
- **Algebraically:** the solution is the  $(x, y)$  that, when plugged in, makes \_\_\_\_\_ equations true.

#### Testing Whether a Point Is a Solution

Plug  $x$  and  $y$  into \_\_\_\_\_ equation. If the point makes \_\_\_\_\_ equations true, it is a solution. If it fails even one, it is not a solution.

#### Example · Is $(2, 5)$ a solution?

EQ 1  $y = 2x + 1$

EQ 2  $y = 3x - 1$

- 1 Test EQ 1 with  $x = 2$ ,  $y = 5$ :  $\underline{\quad} = 2(\underline{\quad}) + 1 = \underline{\quad}$ . True?  $\underline{\quad}$
- 2 Test EQ 2 with  $x = 2$ ,  $y = 5$ :  $\underline{\quad} = 3(\underline{\quad}) - 1 = \underline{\quad}$ . True?  $\underline{\quad}$
- 3 Both true, so  $(2, 5)$  \_\_\_\_\_ a solution.

#### Takeaway

You cannot tell just by looking at a point. Always test \_\_\_\_\_ equations.

### 4 Solving Systems by Substitution

Graphing shows the solution but is slow and hard to read when answers aren't whole numbers. Substitution is algebraic — it always gives an exact answer.

#### Substitution procedure

1. One equation should be solved for a variable (usually  $y = \dots$ ).
2. Substitute the expression into the \_\_\_\_\_ equation.
3. Solve that equation for \_\_\_\_.
4. Plug that value back in to find the \_\_\_\_\_ variable.
5. Write the solution as an \_\_\_\_\_ pair  $(x, y)$  and check.

**Example 1 · Both equations in  $y =$  form**

|      |              |
|------|--------------|
| EQ 1 | $y = 2x + 3$ |
| EQ 2 | $y = -x + 6$ |

- Both equations equal  $y$ , so they equal each other:  $\underline{\hspace{1cm}} = -x + 6$
- Solve for  $x$ . Add  $x$ :  $3x + 3 = 6$ . Subtract 3:  $3x = \underline{\hspace{1cm}}$ . Divide:  $x = \underline{\hspace{1cm}}$ .
- Back-substitute into EQ 1:  $y = 2(\underline{\hspace{1cm}}) + 3 = \underline{\hspace{1cm}}$ .
- Solution: (  $\underline{\hspace{1cm}}$  ,  $\underline{\hspace{1cm}}$  )

**Example 2 · One equation in y = form, one not**

|      |              |
|------|--------------|
| EQ 1 | $y = 3x - 1$ |
| EQ 2 | $2x + y = 9$ |

- EQ 1 is already solved for y:  $y = \underline{\hspace{1cm}}$
- Substitute (  $\underline{\hspace{1cm}}$  ) for y in EQ 2:  $2x + ( \underline{\hspace{1cm}} ) = 9$
- Combine like terms:  $\underline{\hspace{1cm}}x - 1 = 9$ . Add 1:  $5x = \underline{\hspace{1cm}}$ . Divide:  $x = \underline{\hspace{1cm}}$ .
- Back-substitute:  $y = 3(\underline{\hspace{1cm}}) - 1 = \underline{\hspace{1cm}}$ .
- Solution: (  $\underline{\hspace{1cm}}$  ,  $\underline{\hspace{1cm}}$  )

**Practice Problems**

Show your work. Check your answers.

**1. Write the linear equation for this table.**

| x | y  |
|---|----|
| 0 | -4 |
| 1 | -1 |
| 2 | 2  |
| 3 | 5  |

slope  $m = \underline{\hspace{1cm}}$

y-intercept  $b = \underline{\hspace{1cm}}$

Equation:  $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$

**2. Write the linear equation for this table. (no x = 0 row)**

| x | y  |
|---|----|
| 2 | 7  |
| 4 | 13 |
| 6 | 19 |
| 8 | 25 |

slope  $m = \underline{\hspace{1cm}}$

Use  $b = y - m \cdot x$  with (  $\underline{\hspace{1cm}}$ ,  $\underline{\hspace{1cm}}$  ):  $b = \underline{\hspace{1cm}} - (\underline{\hspace{1cm}})(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$

Equation:  $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$

**3. Is (3, 5) a solution to this system? Show your work. Circle YES or NO.**

|      |              |
|------|--------------|
| EQ 1 | $y = 2x - 1$ |
| EQ 2 | $y = -x + 8$ |

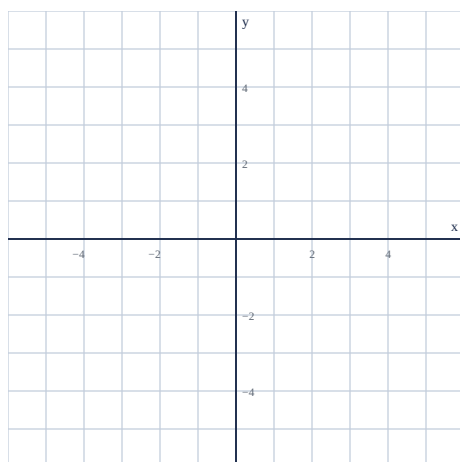
EQ 1 test:  $\underline{\hspace{1cm}} = 2(\underline{\hspace{1cm}}) - 1 = \underline{\hspace{1cm}}$  True or False?  $\underline{\hspace{1cm}}$

EQ 2 test:  $\underline{\hspace{1cm}} = -(\underline{\hspace{1cm}}) + 8 = \underline{\hspace{1cm}}$  True or False?  $\underline{\hspace{1cm}}$

Final answer (circle): YES / NO

**4. Solve this system by graphing on the grid below.**

|      |              |
|------|--------------|
| EQ 1 | $y = x + 1$  |
| EQ 2 | $y = -x + 5$ |



EQ 1 slope = \_\_\_\_, y-int = \_\_\_\_

EQ 2 slope = \_\_\_\_, y-int = \_\_\_\_

Solution: ( \_\_\_\_ , \_\_\_\_ )

**Check:**

EQ 1: \_\_\_\_ = \_\_\_\_ + 1

EQ 2: \_\_\_\_ = -\_\_\_\_ + 5

**5. Solve by substitution.**

|      |              |
|------|--------------|
| EQ 1 | $y = x + 3$  |
| EQ 2 | $y = 2x - 1$ |

Set equal: \_\_\_\_ =  $2x - 1$ Solve for x:  $x = \underline{\hspace{1cm}}$ Find y:  $y = \underline{\hspace{1cm}} + 3 = \underline{\hspace{1cm}}$ 

Solution: ( \_\_\_\_ , \_\_\_\_ )

**6. Solve by substitution. (One equation needs a quick substitution.)**

|      |              |
|------|--------------|
| EQ 1 | $y = 3x + 2$ |
| EQ 2 | $x + y = 10$ |

Substitute ( \_\_\_\_ ) for y in the second equation:  $x + ( \underline{\hspace{1cm}} ) = 10$ 

Combine: \_\_\_\_x + \_\_\_\_ = 10

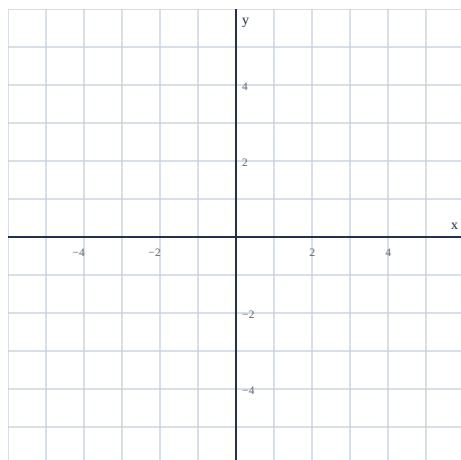
Solve for x:  $x = \underline{\hspace{1cm}}$ Find y:  $y = 3(\underline{\hspace{1cm}}) + 2 = \underline{\hspace{1cm}}$ 

Solution: ( \_\_\_\_ , \_\_\_\_ )

**7. Solve this system by graphing on the grid below.**

EQ 1  $y = 2x - 3$

EQ 2  $y = x + 1$



EQ 1 slope = \_\_\_, y-int = \_\_\_

EQ 2 slope = \_\_\_, y-int = \_\_\_

Solution: ( \_\_\_, \_\_\_ )

**Check:**

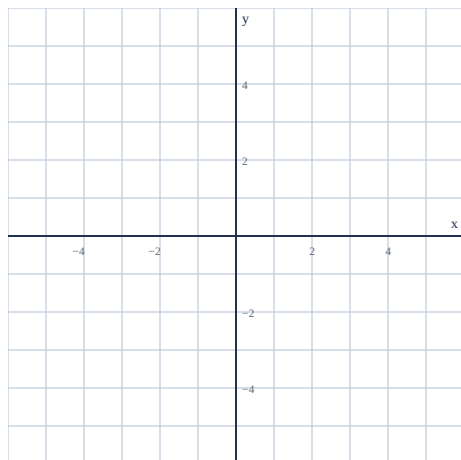
EQ 1: \_\_\_ = 2(\_\_\_ ) - 3

EQ 2: \_\_\_ = \_\_\_ + 1

**8. Solve this system by graphing on the grid below.**

EQ 1  $y = \frac{1}{3}x + 1$

EQ 2  $y = -x + 5$



EQ 1 slope = \_\_\_, y-int = \_\_\_

EQ 2 slope = \_\_\_, y-int = \_\_\_

Solution: ( \_\_\_, \_\_\_ )

**Check:**

EQ 1: \_\_\_ =  $\frac{1}{3}$ (\_\_\_) + 1

EQ 2: \_\_\_ = -\_\_\_ + 5

**9. One, none, or infinitely many? Compare the slopes and y-intercepts first — you do not need to graph these.**a)  $y = 2x + 1$  and  $y = 2x - 4$  → slopes \_\_\_ and \_\_\_, y-intercepts \_\_\_ and \_\_\_ →  
\_\_\_\_\_ solution(s)b)  $y = 3x - 2$  and  $y = -x + 2$  → slopes \_\_\_ and \_\_\_, y-intercepts \_\_\_ and \_\_\_ →  
\_\_\_\_\_ solution(s)c)  $y = -\frac{1}{2}x + 3$  and  $y = -\frac{1}{2}x + 3$  → slopes \_\_\_ and \_\_\_, y-intercepts \_\_\_ and \_\_\_ →  
\_\_\_\_\_ solution(s)**Remember**Different slopes → the lines cross once. Same slope, different y-intercept → parallel, never cross. Same slope *and* same y-intercept → one line written twice.